Emission in the Frequency Range of 2 kHz to 150 kHz caused by electrical vehicle charging

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***Abstract*—This paper studies the conducted emission in the range from 2 kHz to 150 kHz during the charging cycle of electric vehicles. It provides an overview about the necessity of dealing with these emissions and their influencing factors. A test stand which allows measuring the charging of electric vehicles under controlled conditions is described and measurements and analysis methods are presented. The emissions of selected vehicles during the charging cycle at the test stand and in the low voltage grid are shown and compared. The dependencies between the charging state and the main emission frequency as well as between the charging state and the emission level at this frequency are presented.**

***Index Terms*—electric vehicles, test equipment, measurement, switching frequency, harmonic analysis**

I. INTRODUCTION

Standards like IEC 61000-3-2 regulate the harmonic emission of electronic devices in the low frequency range up to 2.5 kHz. The manufacturers of larger equipment like photovoltaic (PV) inverters or charging units for battery electric vehicles [1] (BEV) or plug-in hybrid ones, use technologies with switched power electronics like active power factor correction (aPFC) or PWM controlled rectifiers in order to keep the defined limits and increase the efficiency of the devices [1]. These technologies produce emission in the higher frequency range between 2.5 kHz and 150 kHz. In order to ensure electromagnetic compatibility between the devices and to minimize interferences, emission levels, compatibility levels and immunity levels have to comply with the following concept:

$$\underset{\text{level}}{\text{Emission}} < \underset{\text{level}}{\text{Compatibility}} < \underset{\text{level}}{\text{Immunity}}$$

Whereas the frequency range up to 2 kHz is fully covered by standards, there are only a few standards for the higher frequency range up to 150 kHz. Compatibility levels up to 9 kHz are described only informatively in IEC 61000-2-2, immunity limits are in development (IEC 61000-4-19) and emission limits exist just for a few specific devices and technologies, e.g. induction stoves or mains signalling on power line communication (PLC). There is still a large gap in the standards (no normative compatibility levels, no emission limits for many types of devices), which can result in interferences between devices [2]. During the last years the number of utilities using the frequency range between 2 kHz and 150 kHz for narrowband PLC applications, e.g. meter reading, has been growing. It has to be guaranteed that the "unintentional" emission (e.g. by BEV chargers) is lower than the intentional emission by the PLC in order to ensure its proper functionality.

An increasing number of cases is reported, in which higher frequency emission in a network, e.g. injected by PV inverters or active frontend inverters, lead to malfunction of other devices [3, 4] or erroneous measurements of electronic energy meters [5]. This higher frequency emission can also cause additional losses in the equipment and subsequently lead to additional heating, which can result e.g. in faster aging [6]. Especially devices with a negative impedance frequency characteristic (e.g. shunt capacitors in DC-links of rectifier circuits) are affected. Moreover the higher frequency emission can cause flicker problems when using light dimmers [7] and annoying noise due to mechanical resonances. E.g. one of the analysed vehicles has a switching frequency of about 10 kHz. It generated a well audible noise due to the stimulation of a mechanical resonance within the BEV.

In order to ensure failure-free operation and service of all equipment and devices connected to the grid, it is essential to study the sources, propagation and effects in the higher frequency range. Up to now there are only a few researches known in this area, e.g. [8]. The political framework in Germany promotes both, the increase of PV installations as well as a number of charging BEVs. Both of them use technologies like aPFCs or PWM and emit at higher frequencies. Therefore, detailed analysis of the higher frequency spectra in the grids is needed, i.e. how the emission is generated, how the emission of multiple devices adds, how the emission propagates in the grid and finally what is affected by this emission. This paper focuses on the source characterisation of BEVs in terms of its higher frequency emission.

Because of the present rare spreading of BEVs in Germany, the effects of the increased penetration of charging BEVs on power quality has to be studied by simulations. The simulations have to cover different grids (e.g. rural and urban) with a different penetration of the BEVs. Therefore adequate models for the higher frequency emission of BEVs are needed. The modelling requires measuring data of real BEVs under

[1] In this paper BEV terms all cars, vans etc. with general admission. Lightweight vehicles, electrical bikes etc. are excluded.

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different conditions. In a first step measurements should be carried out in a reproducible, controlled environment. Based on these measurements the respective models are developed and finally verified by measurements in real networks with uncontrolled conditions.

At the beginning the paper describes a test stand, which was used to characterise the higher frequency emission of BEV chargers under laboratory conditions in detail. The next chapter discusses the measurement results from the test stand for different BEVs. Finally the laboratory measurements are compared with measurements of the same BEVs charged in real low voltage grids.

II. Methodology of Analysis

A. Test stand

During measurements in real low voltage grids it is impossible to control the conditions like voltage distortion or grid impedance. Moreover the mentioned factors could change unexpectedly and the results would not be reproducible at all. Hence, a test stand for measuring the charging cycle of BEVs in a defined environment was built.

As shown in fig. 1 it consists of a 3-phase programmable signal source (a), a 3-phase power amplifier of 45 kVA (b) and a variable impedance used as a grid emulator (c). A 10 kW-PV-simulator (d) with three different PV inverters (f) was designed for future measurements of the combined operation of BEVs and PV inverters. Two charging points allow the simultaneous charging of two BEVs. A switch board (f) connects all components and provides interfaces for measurement equipment. In addition to higher frequency emission the test stand is also used to characterise the equipment in terms of low order harmonic emission and unbalance.

There are several standards for connecting BEVs to charging stations. In Germany the most common ones are the so-called *type-1*- and *type-2*-connectors. *SchuKo*, the German standard plug for low voltage devices is used mostly by older vehicles and for fall-back-charging if *type-1* or *type-2*-sockets are not available. All three types are supported by the test stand. But the charging current can only be individually adjusted for each car via a computer interface, if the *type-1*- or *type-2* connectors are used.

The variable impedance consists of several discrete combinable RL-series-elements. It is used for varying the short circuit power of the source. Because it was designed for flicker and low order harmonic analysis, the usability for the higher frequency range is limited. The use of series elements results in too high impedances at higher frequencies. Therefore, these impedance is replaced by external impedances in the future, e.g. as described in IEC 61000-4-7 annex B, which reproduce the harmonic network impedance at higher frequencies more realistically.

Fig. 2 shows a voltage spectrogram of a reference measurement at the test stand without any load. The signal source was programmed to create an ideal sinusoidal voltage with a RMS value of 230 V and a frequency of 50 Hz. Small emissions at 51.2 kHz and 102.4 kHz are visible in the output

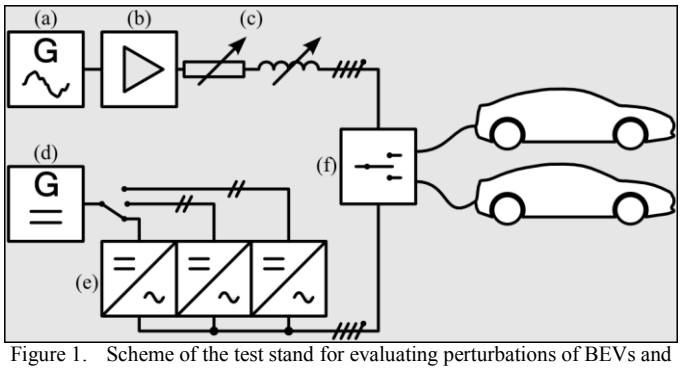


Figure 1. Scheme of the test stand for evaluating perturbations of BEVs and photovoltaic inverters. (a) signal generator, (b) 3 power amplifiers, (c) changeable impedance, (d) PV simulator, (e) PV inverters, (f) switch board

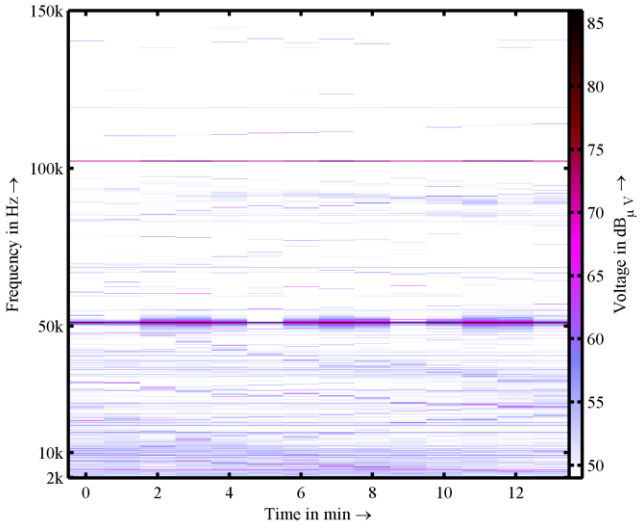


Figure 2. Analysis of the no-load behavior of the test stand.

voltage. These are the switching frequency and its harmonic caused by signal source itself. The oscillation of the spectra and levels around 51.2 kHz is an aliasing effect of the PWM-control of the amplifiers generated by the FFT analysis and aggregation in time (cp. chapter II-D). The small emission lines originate from the digital-analogue-converter and are considerably damped by connecting any load to the amplifier. These "idle emission" can be easily identified and will be excluded from the analysis.

B. Analysed vehicles

Until now the charging characteristic of 20 different types of BEVs was measured:

- 6 in public low voltage networks only
- 8 at the test stand only
- 6 both in low voltage networks and in the laboratory.

One aspect of the paper is the comparison of emission

Table 1. Overview of BEV's and their basic Parameters

Vehicle	Plug	Rectifier topology	Max. charging current
BEV1	Type-2	3x single phased	31.4 A
BEV2	Type-2	2x single phased	14.5 A
BEV3	Type-1	single phased	16.0 A
BEV4	Type-1	single phased	14.6 A
BEV5	SchuKo	single phased	15.9 A
BEV6	Type-2	single phased	15.4 A

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under laboratory conditions and real grid conditions. Therefore this paper only focuses on the six BEVs measured in grid and laboratory. Tab. 1 contains an overview of them, presenting their connection type and maximum charging current.

C. Impact factors and test setup

It cannot be assumed that the emission in the range of 2 kHz to 150 kHz is constant during the whole charging cycle of a BEV. Therefore at the beginning of the study possible factors with impact on the emission were identified. The factors can be classified into grid depending and vehicle depending ones:

Vehicle	• Rectifier topology (including aPFC or PWM modes) and control • Battery charging state • Charging current limitation
Grid	• Low frequency harmonic distortion (up to 2 kHz) • Higher frequency harmonic distortion (2 kHz to 150 kHz) • Voltage unbalance • Voltage magnitude • Power frequency • Grid impedance

The rectifier topology and control are determined by the manufacturer design and cannot be changed. The charging current can be limited in discrete steps by the charging station, if *type-1* or *type-2* connectors are used. The impact factors of the grid can be individually adjusted by the controllable source used in the test stand.

All changeable factors are covered during the measurements at the test stand. At the beginning the whole charging cycle of each vehicle is measured using ideal conditions (sinusoidal, no unbalance, frequency of 50 Hz, RMS-value of 230 V, no impedance) in order to analyse the impact of charging state and rectifier topology (initial test) on the higher frequency emission. In a subsequent measurement the impact of the grid-dependent factors is analysed. Therefore, all factors are varied according to predefined schemes within the limits according to IEC 61000-2-2. The measurements are repeated for different impedances. If a *type-1* or a *type-2*-connector is used, a current limit in discrete steps is applied while the car is charged another time under ideal conditions.

This paper presents the results of the first step of the analysis, namely the impact of the topology and the charging state. The impact of the other parameters will be presented in future papers.

D. Measurement and data handling

The data was recorded with power quality analysers at a sampling rate of 1 MS/s and a resolution of 16 bits. For reducing the amount of data, the raw data was recorded every minute for a variable interval between 1 s and 3.1 s. In order to prevent leakage effects, a digital high pass filter (elliptic, 3rd order, 2 kHz pass band) was applied to the stored datasets. For the higher frequency analysis the data of each measurement interval was split into 200 ms blocks according to IEC 61000-4-7 (see fig. 3). Afterwards an FFT was used to obtain the frequency spectra in 5 Hz bands. The spectra of all blocks of each measurement interval were averaged using RMS method.

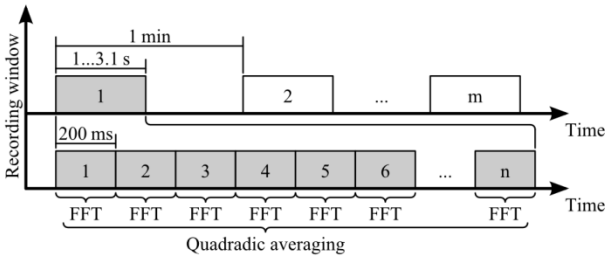


Figure 3. Measuring interval and splitting of the data for FFT analysis

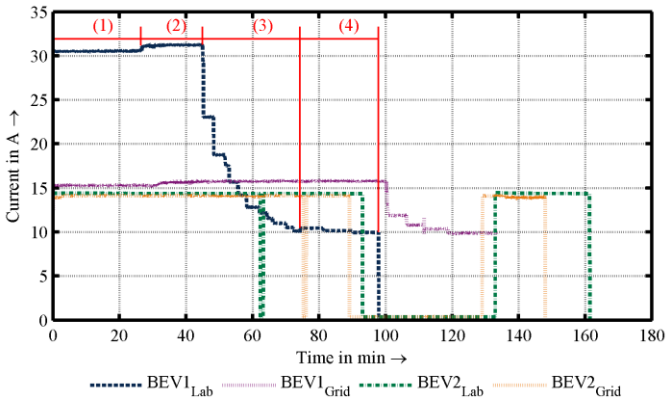


Figure 4. Currrent in L1 during the charging cycle of *BEV1* and *BEV2* in the laboratory and in the grid.

All higher frequency spectral levels are presented in the logarithmic unit $dB_{\mu V}$ ($dB_{\mu A}$), where 120 $dB_{\mu V}$ correspond to 1 V and 0 $dB_{\mu V}$ to 1 µV (analog for currents). One decade is represented by 20 dB (e.g. 100$dB_{\mu V}$ ≙ 100 mV).

III. Results of Measurements

A. Charging cycle characteristic

Usually a charging cycle consists of different states [9]. This can be seen in fig. 4, which shows exemplarily the charging cycle of *BEV1* and *BEV2* for the grid measurement and the measurement at the test stand. During the grid measurement the charging current of BEV1 was limited to 16 A due to the rating of the used charging station. Therefore, the grid measurement of *BEV1* is not completely comparable with the laboratory measurement where no current limitation was set. The charging cycle of *BEV1* consists for both measurements of four different states which are indicated in fig. 4 for the laboratory measurement. In the first two states the current is kept nearly constant due to the internal limitation of the vehicle (constant current charging). In the remaining two states the current follows an exponential curve, which corresponds to a constant voltage charging of the internal batteries. In contrast, the charging current of *BEV2* is constant for most of the time, but interrupted for several minutes.

Each different charging state including the pauses, which is not negligibly short, may require different parameters for a simulation model (see analysis of frequencies and levels below). Therefore, the moment of transitions between the

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states need to be analysed individually for every BEV. Indications for the transitions may be the passed time or the charging state of the battery. For example, for BEV1 the following was already analysed:

- State 1 → State 2:
  Occurs about 30 min after start of charging cycle (independent of charged energy, cp. fig. 4). The small difference in time can be neglected.
- State 3 → State 4:
  For both cycles at a transmitted energy of $E_{tx} \approx 0.95 \cdot E_{max}$ respectively about 25 min before end of charging.

There is no unique behaviour for all BEVs. The number of states and their duration are usually different between the BEVs. Some vehicles have a not negligible duration of the constant voltage charging state (e.g. *BEV1*, *BEV3*). For other vehicles the constant voltage charging state lasts very shortly (e.g. *BEV4*, *BEV6*). Others have only states with constant current charging (*BEV2*, *BEV5*). Tab. 2 summarizes the charging cycle characteristics of the analysed BEVs.

B. *Switching frequency*

Concerning the switching frequency the BEVs behave as differently as they do in their charging cycle characteristic. This is clearly shown by the difference between fig. 5 and fig. 6. They present the spectrograms of the charging current of *BEV1* and *BEV5*. The frequency range in both figures is limited to 50 kHz because no significant emission is visible above.

Fig. 5 (*BEV5*) shows that the switching frequency can change abruptly at the transition between two charging states. The emission is spread over a wide frequency range, which makes it difficult to estimate the main switching frequency. In contrast, for *BEV1* (fig. 6) a distinctive switching frequency spread over a narrow frequency range at about 10 kHz can be identified. A second, wider band at around 6.5 kHz is superimposed. It decreases rapidly after the transition from constant current charging to constant voltage charging. *BEV3* changes its switching frequency continuously during the charging cycle. The frequency shows an inverse exponential increase over the time. In contrast, *BEV2* and *BEV4* have no identifiable switching frequency in the range between 2 kHz and 150 kHz. The switching frequency behaviour of the analysed BEVs can be classified into the scheme according to tab. 3.

Changes from narrow band emission to wide band emission and vice versa could not be observed. Analogue to the PV inverters (e.g. [4]) multiples (harmonics) of the main

TABLE 2. SUMMARY OF CHARGING CYCLE CHARACTERISTIC

Vehicle	Charging States[1]	Number of states[2]
BEV1	I1, I2, U1, U2	4
BEV2	I1, interrupted	2
BEV3	I, U	2
BEV4	I, (U)	1
BEV5	I1, I2	2
BEV6	I, (U)	1

[1] Const. I resp. const. U charging; brackets imply negligible short duration
[2] Different states needed for modelling

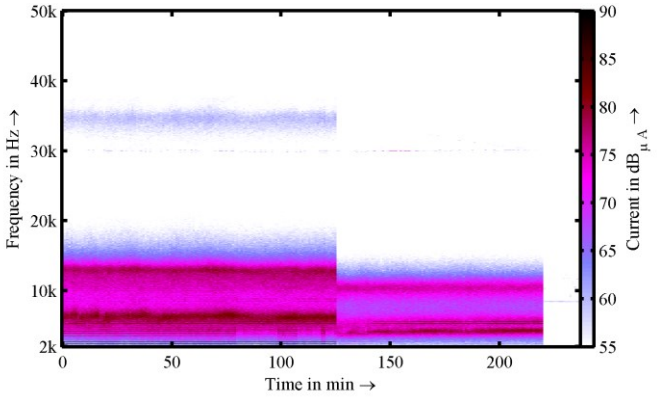


Figure 5. Spectrogram of the current in phase L1 during the charging cycle of *BEV5* at the test stand

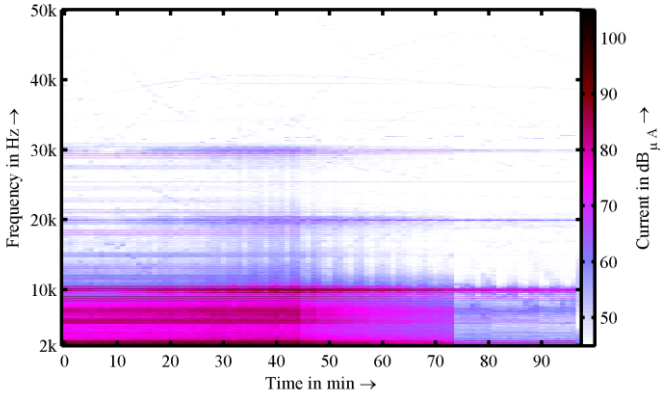


Figure 6. Spectrogram of the current in phase L1 during the charging cycle of *BEV1* at the test stand.

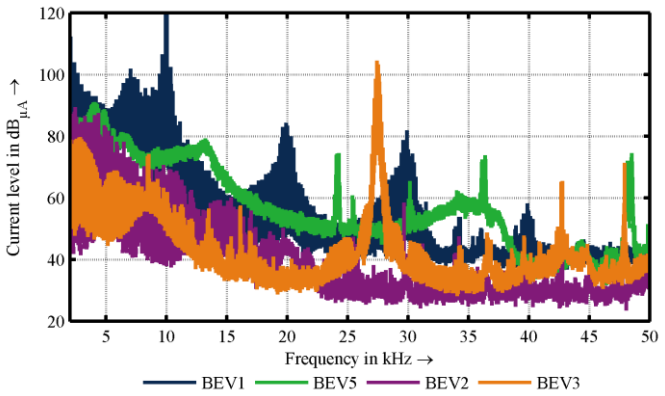


Figure 7. Current spectra of several BEVs during constant current charging at the test stand

TABLE 3. CLASSIFICATION OF SWITCHING FREQUENCY BEHAVIOUR

Bandwidth	Switching frequency
none	none
narrow	constant
	variable (discrete)
	variable (continuous)
wide	constant
	variable (discrete)
	variable (continuous)

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emission band at the switching frequency are clearly visible in the frequency spectra (fig. 7).

Tab. 4 summarises the characteristics of all vehicles in terms of the switching frequency. The spectral distributions of *BEV1*, *BEV2*, *BEV3* and *BEV5* are shown in fig. 7. The spectra are calculated using one measurement interval during the first constant current charging state. *BEV3* is an example for a vehicle with a very narrow emission band whereas the emission of *BEV5* has a smooth shape with small, superimposed peaks at the switching frequency and its harmonics. This is hardly recognisable in the spectrogram in fig. 6. The comparison between the spectrograms and the spectral distribution plots illustrates that both ways of presentation are required for a comprehensive analysis.

C. *Emission level at switching frequency*

Beside the switching frequency, the emission level at switching frequency is another important parameter to characterise the emission of BEVs. The emission level represents the energy which is injected into the grid and may disturb other devices.

As shown above the emission at switching frequency spreads over a certain frequency range (bandwidth at switching frequency), which can be narrow or wide. According to [4] 600 Hz-bands centred around the switching frequency would catch more than 99% of the emission of PV inverters. This approach cannot be applied to BEVs due to the possibility of wide band emission. Due to the lack of a better alternative and considering the resulting error, a bandwidth of 200 Hz was chosen for the analysis in this paper. This corresponds to the recommendation of IEC 6100047, annex B. A proper method to aggregate narrow and wide band emission has to be developed in the future.

Fig. 8 shows the time characteristic of the emission level at the switching frequency of the BEVs with a significant switching frequency. For *BEV1* and *BEV5* the emission level depends on the charging state (constant behaviour followed by exponential behaviour for *BEV1* and sudden decrease of emission level at the moment of the switching frequency change for *BEV5*). *BEV3* and *BEV6* show almost constant emission levels which are independent of the charging state.

Tab. 5 contains an overview of the maximum emission level of each vehicle. These levels were all measured during the first constant current charging state of each vehicle. The ongoing analysis of the impact of grid parameters is focused on this first constant current charging state in order to identify if certain conditions may cause higher emission levels than the ones measured under ideal conditions.

Fig. 9 compares the emission at the switching frequency of all vehicles for the grid measurement and the laboratory measurement for the whole charging cycle. For *BEV1*, *BEV5* and *BEV6* the difference between both measurements is small. The levels of *BEV3* differ considerably between the grid and the laboratory measurement. The comparison shows that a general statement about the influence of the grid parameters on the charging cycle of all BEVs is not possible. However, for most of the analysed vehicles (except for *BEV3*) the difference between grid and laboratory measurements is not too large.

TABLE 4. SUMMARY OF SWITCHING FREQUENCY

Vehicle	Bandwidth	Frequency behaviour	Switching Frequency
BEV1	narrow	constant	10.03 kHz
BEV2	-	-	-
BEV3	narrow	variable (continuous)	≈ 28 kHz (inverse exponential)
BEV4	-	-	-
BEV5	wide	variable (discrete)	6.55 kHz, 5.2 kHz
BEV6	narrow	constant	11 kHz

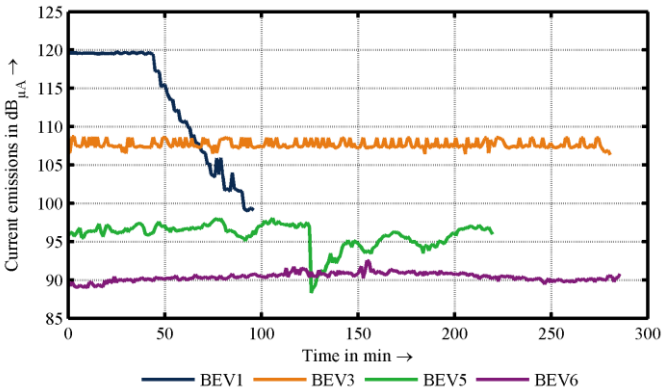


Figure 8. Current emission level of *BEV1*, *BEV3*, *BEV5* and *BEV6* at switching frequency during the complete charging cycle (laboratory measurements)

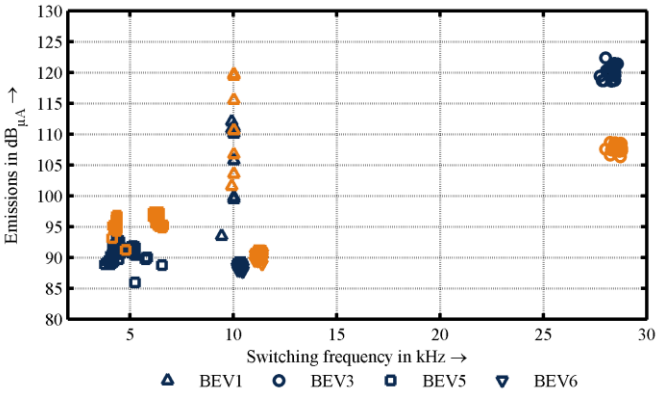


Figure 9. Emission levels at switching frequency of the BEVs. Dark colour represents the grid measurement, bright colour the laboratory measurement

TABLE 5. OVERVIEW OF EMISSION LEVELS AT SWITCHING FREQUENCY BASED ON 200-HZ-BANDS

Measurement	Max. current emission level			
	BEV1	*BEV3*	*BEV5*	*BEV6*
Grid	112.3 $dB_{\mu A}$	122.4 $dB_{\mu A}$	93.1 $dB_{\mu A}$	89.6 $dB_{\mu A}$
Laboratory	119.8 $dB_{\mu A}$	108.9 $dB_{\mu A}$	98.0 $dB_{\mu A}$	92.5 $dB_{\mu A}$

[10] provides immunity levels for electricity meters. In the frequency range from 2 to 30 kHz these levels amount 2 A or 126 $dB_{\mu A}$. The measured emission of the analysed vehicles (tab. 5) does not exceed the immunity levels, but the highest measured level is only 4 $dB_{\mu A}$ lower. The maximal voltage level arose while charging *BEV3*. During the grid

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measurements a maximum level of 122.5 $dB_{\mu V}$ was observed. According to tab. 1 in [11] for narrow band PLC at 28 kHz a maximum transmission level of about 127.3 dBµV is allowed. The measured level is only slightly lower than the maximum transmission level. However, a disturbance-free PLC communication requires a minimum difference between the unintentional and the intentional emission. Furthermore, the emission levels of the vehicles should be considerably lower than the maximum transmission level.

It should be noted that the measurement method can have a significant impact on the results [12]. For a precise comparison the methods for the analysis of the measurement data have to comply with the measurement methods defined in the standard. All calculations in this paper were performed using the RMS value of 200 Hz-bands. While the values in [10] represent test values at a single frequency, [11] specifies a spectrum analyzer with a peak detector and a resolution bandwidth of 100 Hz.

D. Basic Modelling Approach

A first simulation model should include the switching frequency and its harmonics derived from lab measurements. The grid-dependent impact factors are still neglected at this stage. In the first step only the first charging state (constant current) is considered, because the highest emission occurs usually during this state.

For future models the stepwise inclusion of grid-dependent impact factors as well as different charging states can be considered.

IV. Conclusion

This paper provides a brief overview of the higher frequency emission in the range between 2 kHz and 150 kHz caused by BEV chargers. A test stand is used for the detailed characterisation of the emission. For verification purposes the analysed vehicles were also measured in real low voltage grids.

The emission depends on several vehicle-dependent and grid-dependent impact factors. This paper studies the emission behaviour in regard to the charging state characteristics for six different BEVs. For each BEV the switching frequency and the emission level at the switching frequency were analysed.

Not all BEVs show a significant switching frequency and the switching frequency is neither constant during the charging cycle nor similar between the BEVs. The emission levels depend on the charging topology of each vehicle and the charging state. The higher frequency emissions of the vehicles were found in the frequency range of about 3 kHz to 29 kHz. The emission levels range from 85 $dB_{\mu A}$ to 125 $dB_{\mu A}$, which is still below the immunity levels for electricity meters. The measured voltage levels are also lower than the maximum transmission level of PLC signalling. But for one BEV the difference between both levels is small. It is important to respect a minimum difference in order to ensure an error-free PLC communication. This should be considered, e.g. in the future standardisation activities.

For a first model it is recommended to focus only on the switching frequency and the charging state with the highest emission.

The dependency of higher frequency emission on the grid-specific impact factors introduced in chapter II is currently analysed. Based on this results possible improvements and generalisations of the existing models are assessed.

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